

PROBLEM STATEMENT:

How can we use historical sales-related information such as advertising, price, discount, number of customers, and month to predict future sales accurately, and which factors have the greatest influence on sales?

TITLE:

Sales Prediction Using Machine Learning Techniques

OBJECTIVE:

To analyze sales data, visualize important patterns, train a machine learning model, predict sales values, evaluate model performance, and identify the most important sales-related features.

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

np.random.seed(42)
n = 1000

advertising = np.round(np.random.uniform(5, 100, n), 2)
price = np.round(np.random.uniform(50, 500, n), 2)
discount = np.round(np.random.uniform(0, 30, n), 2)
customers = np.random.randint(50, 1000, n)
month = np.random.randint(1, 13, n)

sales = (120 + 5.2 * advertising + 0.75 * customers
        - 0.45 * price + 8.0 * discount
        + 20 * np.sin(month / 12 * 2 * np.pi)
        + np.random.normal(0, 120, n))
sales = np.round(np.maximum(sales, 50), 2)

df_sales = pd.DataFrame({
    'advertising': advertising,
    'price': price,
    'discount': discount,
    'customers': customers,
    'month': month,
    'sales': sales
})

print("First 5 rows of Sales data:")
print(df_sales.head())

print("\nMissing Values:")
print(df_sales.isnull().sum())

print("\nSummary Statistics:")
print(df_sales.describe())

sns.histplot(df_sales['sales'], kde=True)
plt.title('Sales Distribution')
plt.show()

sns.scatterplot(x='advertising', y='sales', data=df_sales)
plt.title('Advertising vs Sales')
plt.show()

X = df_sales[['advertising', 'price', 'discount',
               'customers', 'month']]
y = df_sales['sales']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

rf_model = RandomForestRegressor()
```

```
        n_estimators=100, random_state=42)

rf_model.fit(X_train, y_train)
sales_pred = rf_model.predict(X_test)

print("\nRandom Forest Sales Prediction Results:")
print("MAE:", mean_absolute_error(y_test, sales_pred))
print("MSE:", mean_squared_error(y_test, sales_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, sales_pred)))
print("R2 Score:", r2_score(y_test, sales_pred))

importance = rf_model.feature_importances_
feature_names = X.columns

sns.barplot(x=importance, y=feature_names)
plt.title('Feature Importances (Sales Prediction)')
plt.show()
```

OUTPUT:

First 5 rows of Sales data:

advertising	price	discount	customers	month	sales
40.58	133.31	7.85	932	8	1206.29
95.32	293.86	7.41	362	11	873.1
74.54	442.83	27.19	889	8	1343.24
61.87	379.5	7.49	568	8	578.49
19.82	412.95	8.16	255	12	129.87

Missing Values:

```
advertising    0
price          0
discount       0
customers      0
month          0
sales          0
dtype: int64
```

Summary Statistics:

	advertising	price	discount	customers	month	sales
count	1000.00	1000.00	1000.00	1000.00	1000.00	1000.00
mean	51.57	278.16	15.07	536.57	6.62	780.86
std	27.75	131.49	8.72	276.82	3.43	294.17
min	5.44	51.45	0.00	52.00	1.00	50.00
25%	27.42	158.48	7.84	296.75	4.00	566.24
50%	52.20	283.43	15.02	540.50	7.00	789.24
75%	75.71	392.21	22.77	775.25	10.00	998.96
max	99.97	499.74	29.93	999.00	12.00	1728.23

Sales Statistics:

Minimum Sales: 50.00

Maximum Sales: 1728.23

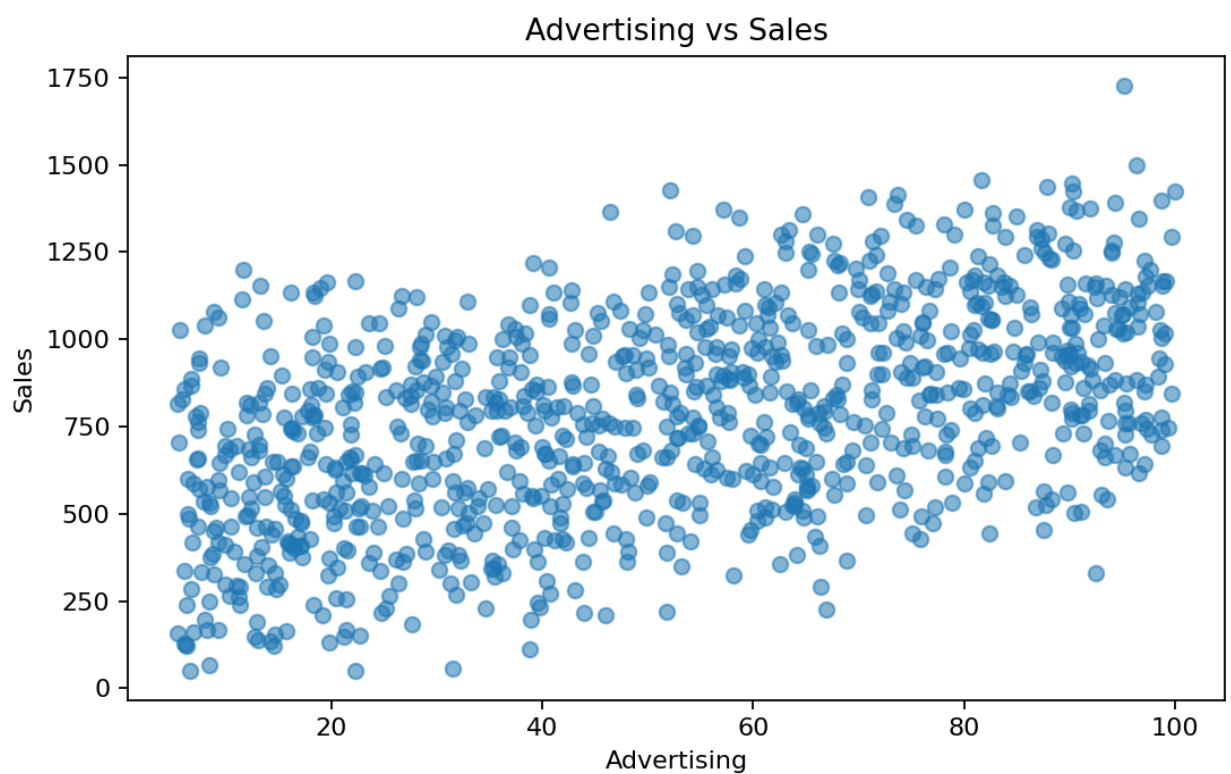
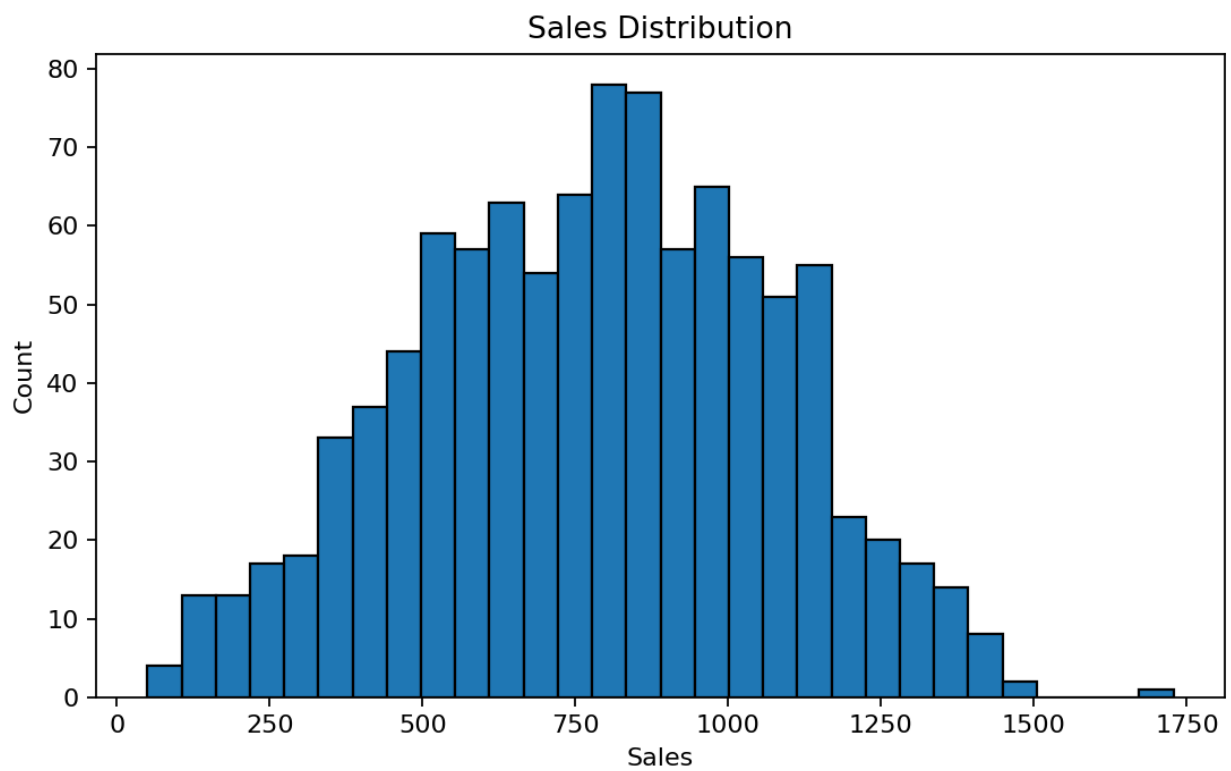
Average Sales: 780.86

Median Sales: 789.24

Total Records: 1000

The dataset contains no missing values. The sales values are used as the target variable, while advertising, price, discount, customers and month are used as prediction features.

OUTPUT:



RANDOM FOREST SALES PREDICTION RESULTS:

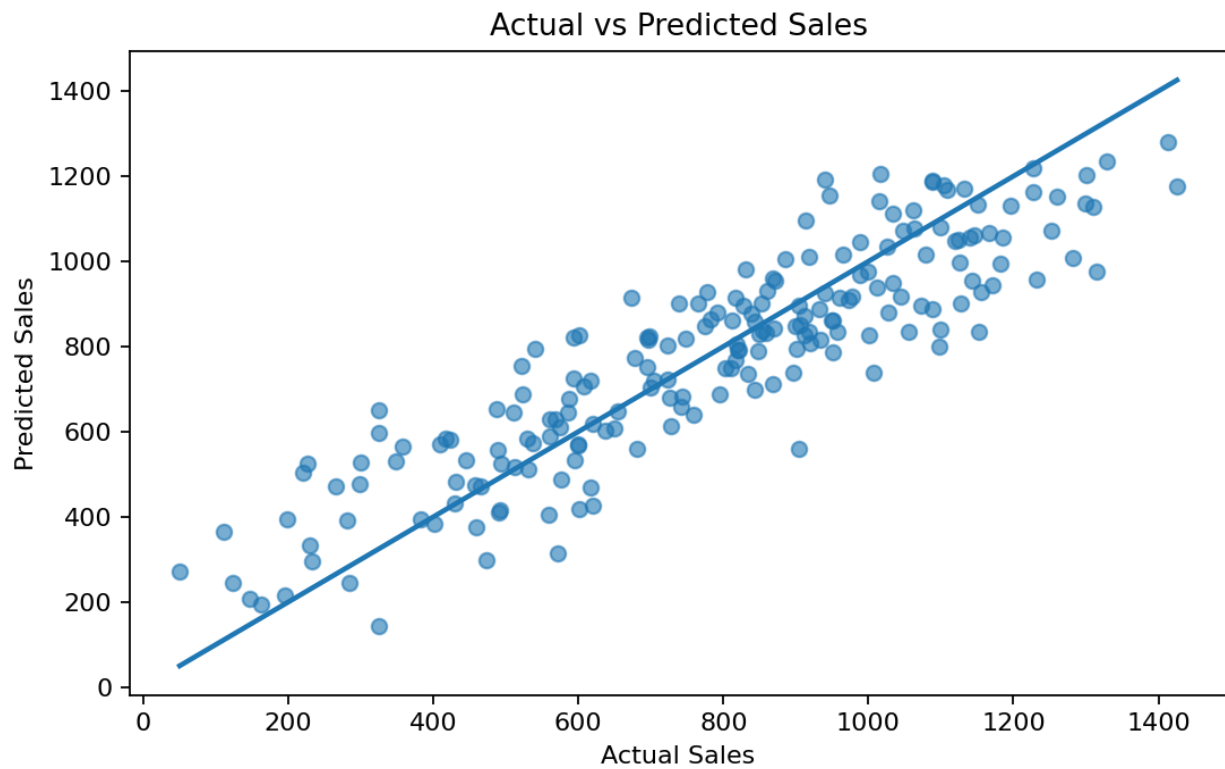
MAE : 110.68

MSE : 18730.33

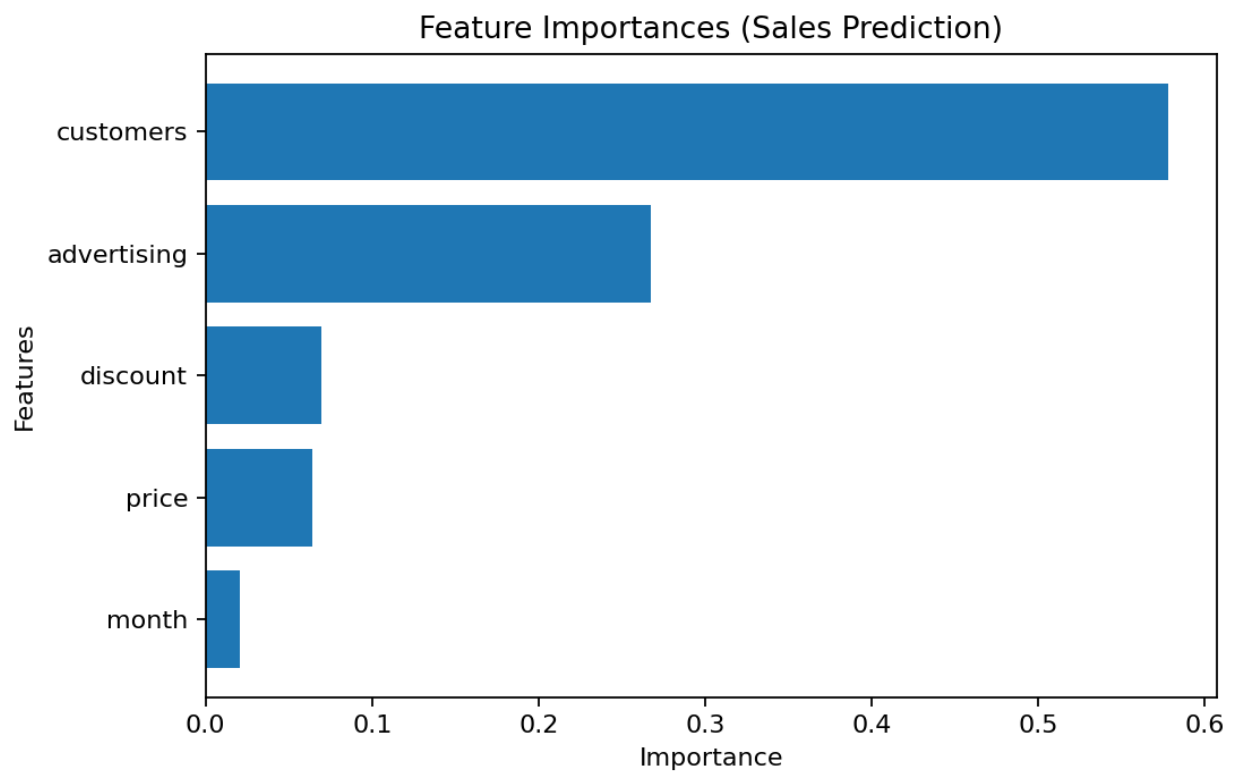
RMSE : 136.86

R2 SCORE : 0.7925

Actual vs Predicted Sales:



FEATURE IMPORTANCES (SALES PREDICTION):



The feature-importance graph shows the relative contribution of the input variables used by the Random Forest model. It helps identify which available business variables are most useful for predicting sales.

CONCLUSION:

The Sales Prediction project demonstrates a complete data science workflow: data preparation, exploratory data analysis, visualization, model training, prediction, evaluation, and feature-importance analysis. The Random Forest Regression model predicts numerical sales values from multiple business inputs.

FUTURE ENHANCEMENTS:

1. Use a real company sales dataset.
2. Add product category, region, season, holidays and customer segments.
3. Compare Random Forest with other regression algorithms.
4. Deploy the model as a web application or dashboard.
5. Connect the project to live business data.